Field of Dental Medicine

Biomaterials II

Lecture : (Dental Wax)

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Date : 12/ 4 /2023

Dental Waxes

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References

1.Dental materials foundations and applications 11ed.

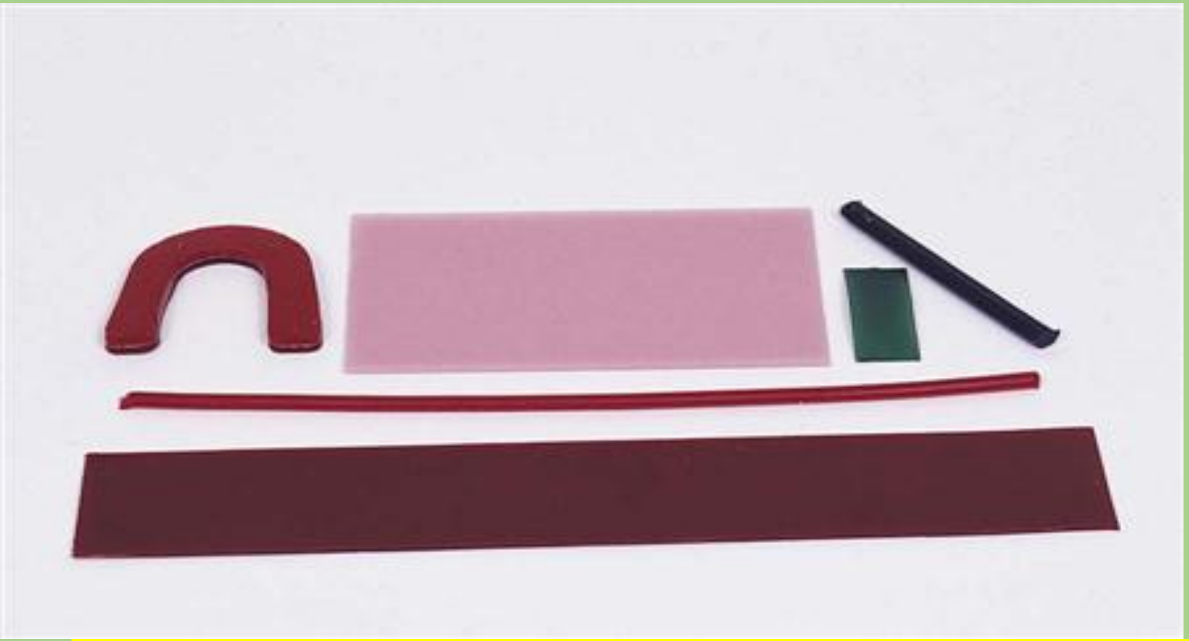
2. pocket dentistry website

3- Cairo university –Faculty of dental medicine, Textbook of dental Biomaterials Department.

Introduction & composition

- Waxes are used in many aspects of dentistry either in the clinic or in the laboratory
- Dental waxes are always a **mixture** of various components.
- Components could be: - **natural waxes from minerals, plants, or animals.**

  - **they may be synthetic waxes**



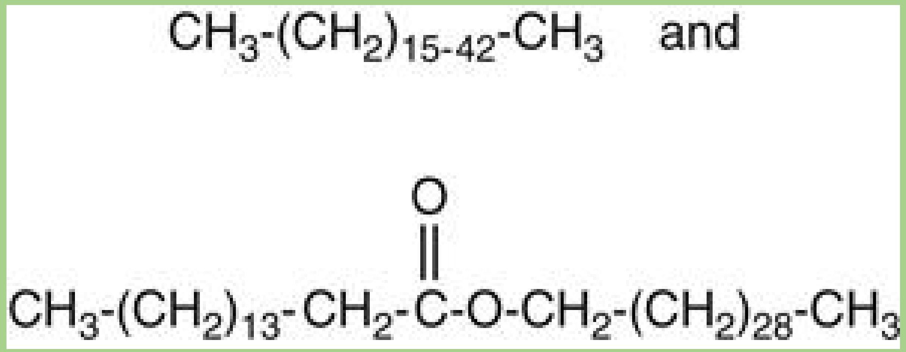
https://pocketdentistry.com/18-dental-waxes/

- **Gums, fats, fatty acids, oils, and various resins** may be added to modify and control the properties.

- **Pigments** may be added for color
- **Fillers** may be added

- Waxes are **organic (hydrocarbons) with high molecular weight**

- **Long chains contain carbon, hydrogen and oxygen**



The top structure is a hydrocarbon (from paraffin),and the bottom structure is a high molecular weight ester. The parentheses in the structures indicate repeating units of carbon and hydrogen.

Types & General Properties

- The **long chains make waxes flexible** at room temperature or sticky as solids or as liquids.

- Properties such as hardness, melting range, and flow depend on the **amounts and ratios** of the different waxes and their molecular structure.

Dental waxes

Pattern wax : casting wax, baseplate wax and inlay wax

Processing wax: beading wax , boxing wax, utility and sticky wax

- **<u>1-Melting range & Thermal properties:</u>**
- 1- **No definite melting point**, because they are a mixture of different components
- 2- Waxes have **melting range**
- **3-At the low end of the range**, some but not all of the components melt, which causes the wax (still solid) to flow much more.
- **4-As the temperature increases** through the melting range, more of the components melt and the wax flows more and more, then all components become a liquid.

- Solid-solid transition temperature is the temperature below melting range

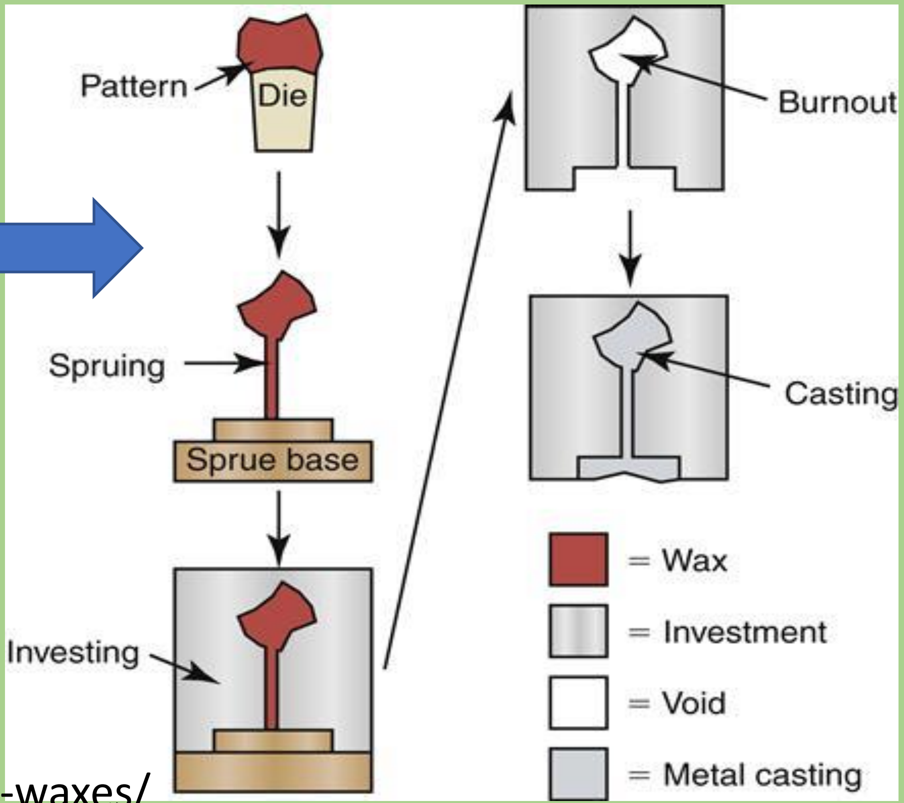
- Wax at solid-solid transition temperature will acquire flow and ductile properties

- You can manipulate and carve wax at solid-solid temperature satisfactory without flaking or tearing

2-Excess Residue:

- Because wax patterns used in the lost-wax technique are melted or burned or ignited to remove them from the casting mold, the wax must not leave a residue or remnants, which can affect the quality of the final restoration.

Lost-wax technique.



https://pocketdentistry.com/10-waxes/

3-Flow:

Flow is the change in **shape under an applied force**.

It is caused by the **slippage of the long-chained wax molecules** over each other.

Flow is highly dependent on **temperature and time**.

At low temperatures, waxes nearly do not flow at all, but as the **temperature approaches the melting range of the wax**, the flow increases dramatically

For pattern waxes, flow is generally not required at room or mouth temperature, because it results in a permanent distortion of the wax pattern.

For boxing and utility processing waxes, flow is a highly desirable property because these waxes need to be pliable at room temperature.

Flow of wax.

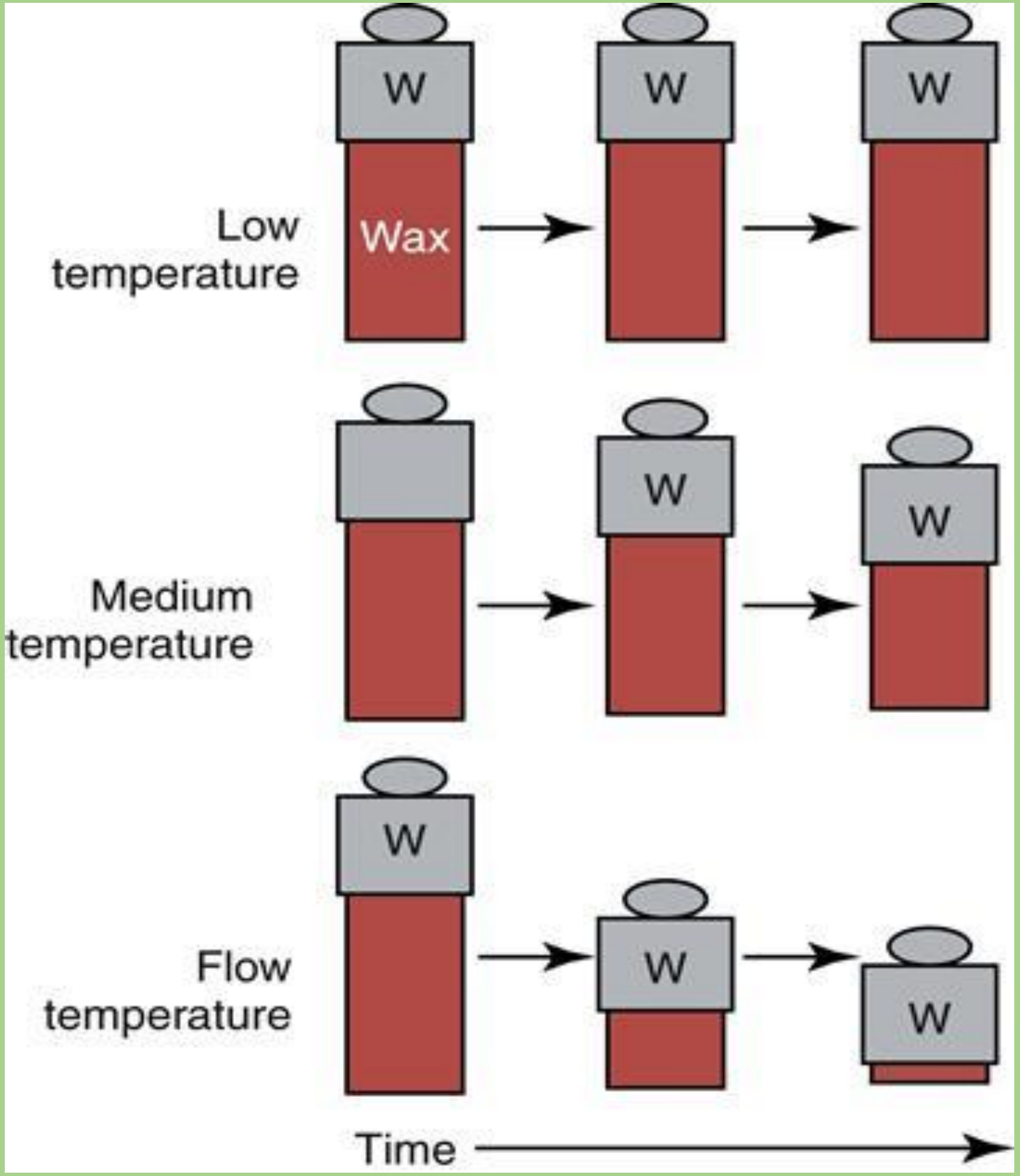
Flow is a function of time and temperature.

If a small weight, W, is placed on top of a cylindrical wax sample (shaded area), no change in height occurs if the wax is at a low temperature relative to its flow temperature (top row).

As the temperature approaches the flow temperature, some change in dimension occurs over time, medium temperature (second row).

At the flow temperature, large changes in dimension occur, flow temperature (bottom row).

In all cases, the amount of flow is time dependent and usually is expressed as a percentage of the original height.



- **Flow = plastic deformation = Time, force and temperature dependent**
- ***Flow*** is **low at temp. under solid-solid transition temperature**
- It increases as temperature increases above transition temperature
- Flow ***gives wax ductile properties***
- Wax ductility increases as temperature increases
- You should manipulate wax (carving , shaping etc….) in the solid-solid transition temperature ***not higher as the wax will be molten not lower as the wax would be too hard and will be liable to flake and cracking.***

- **4-Thermal Expansion:**

- When waxes are heated, **they expand significantly.**
- This expansion can be quantified as a percentage of the original dimension of the specimen
- In general, ***waxes have the highest coefficients of thermal expansion of any dental material***
- **In pattern waxes, the thermal expansion is so critical, as small changes in temperature can cause sufficient changes in dimensions to make the pattern inaccurate.**

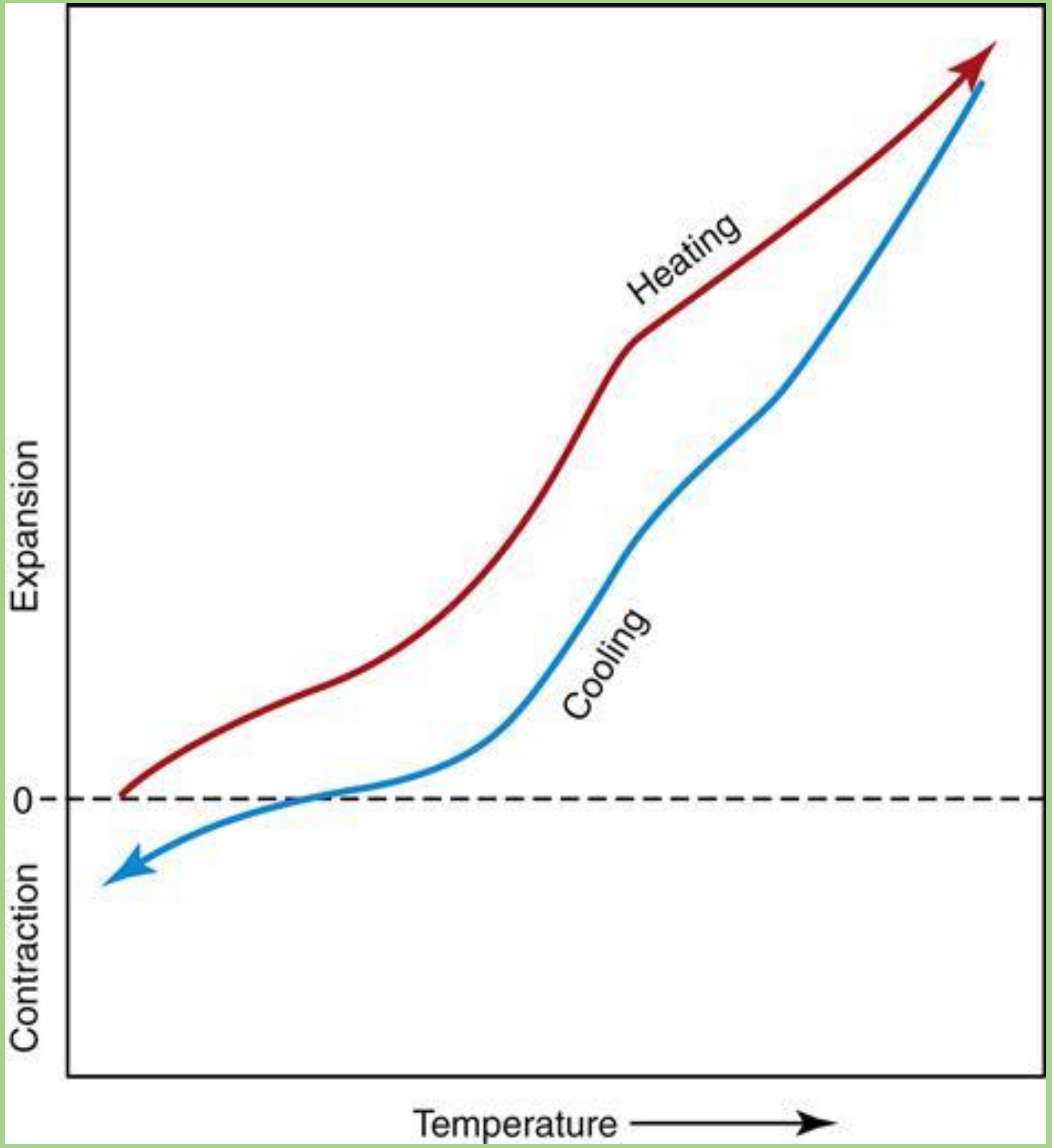
- **Thermal expansion of wax.**

- As wax is heated, a significant expansion takes place but **<u>may not be linear with the increase in temperature.</u>**

- As the wax cools, contraction occurs

<u>but not reversibly.</u>

So, a wax pattern that goes through a heating-and-cooling cycle may have distorted dimensions



- **5-Residual Stress:**
- Residual stress is the stress remaining in wax as a result of **manipulation** during heating, cooling, bending, carving, or other manipulations.
- Manipulation of wax puts molecules of the wax into positions **that they hate** but cannot change because of their solid state, causing generation of **residual stresses**.
- These stresses are released **as the temperature of the wax increases**, and the wax molecules can move more freely.

Residual stresses

- A pattern wax is used to create a model of a dental restoration that will be cast using the **lost-wax technique**
- **The release of residual stresses** at higher temperatures causes an irreversible changes and deformation that can destroy the accuracy and fit of a wax pattern.

https://www.researchgate.net/publication/215935405_Waxing_Techniques_to_Develop_Proper_Occlusal_Morphology_in_Different_Occlusal_Schemes/figures

- **HOW to overcome this problem?:**
- 1- **preventing residual stresses from forming :**waxes **should not** be carved or burnished **at cold temperatures** below their solid-solid transition temperature. **So Wax patterns are carved with warm instruments, and melted wax is added in small increments (increment by increment) to prevent rapid sudden or uneven cooling, to prevent residual stress.**
- 2- **To prevent the release of stress already created: A-**wax should not be subjected to temperature changes and not to be stored at high temperatures
- **B - The time between finishing and investing the pattern should be minimized (about 10 minutes)** because longer storage times allow time for release of stresses

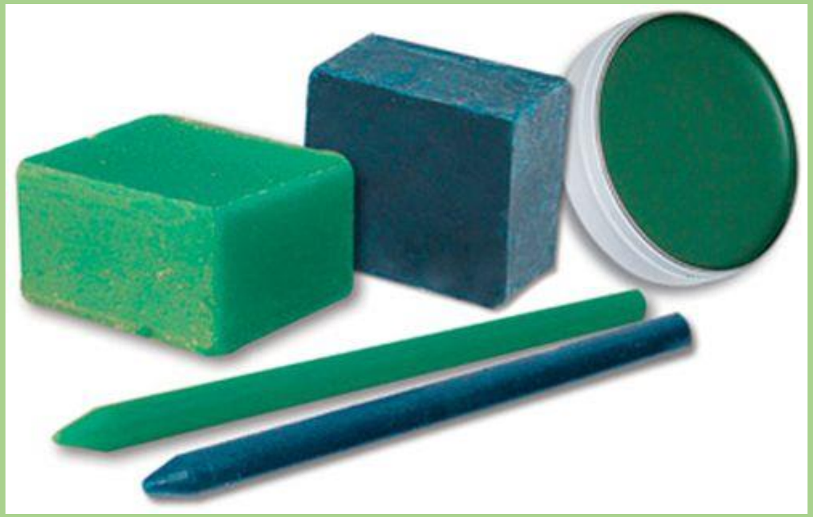
I- PATTERN WAXES

- Pattern waxes are used to create a model of a dental restoration such as a crown or partial denture

Wax pattern of a crown restoration attached to a sprue and ready for investing. The crown pattern is upside down to facilitate placement of the sprue and casting.

https://pocketdentistry.com/10-waxes/

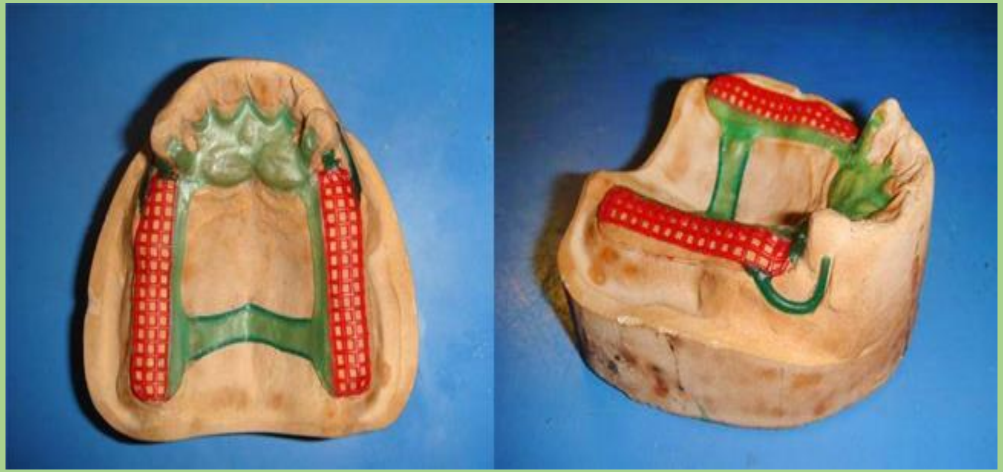
- <u>**A-Inlay Wax**</u>
- Inlay waxes generally are **used to fabricate wax patterns for crowns, inlays, or bridges.**
- These waxes generally are available in **round sticks (7.5 cm long and 6 mm in diameter) of several colors such as red, yellow, blue, and green.**
- The composition of inlay wax is **complex**, and it may contain five or six different waxes, such as **paraffin, carnauba, ceresin, and beeswax.**
- **Paraffin and ceresin are mineral waxes**
- **carnauba is a plant wax, and beeswax is an insect wax.**

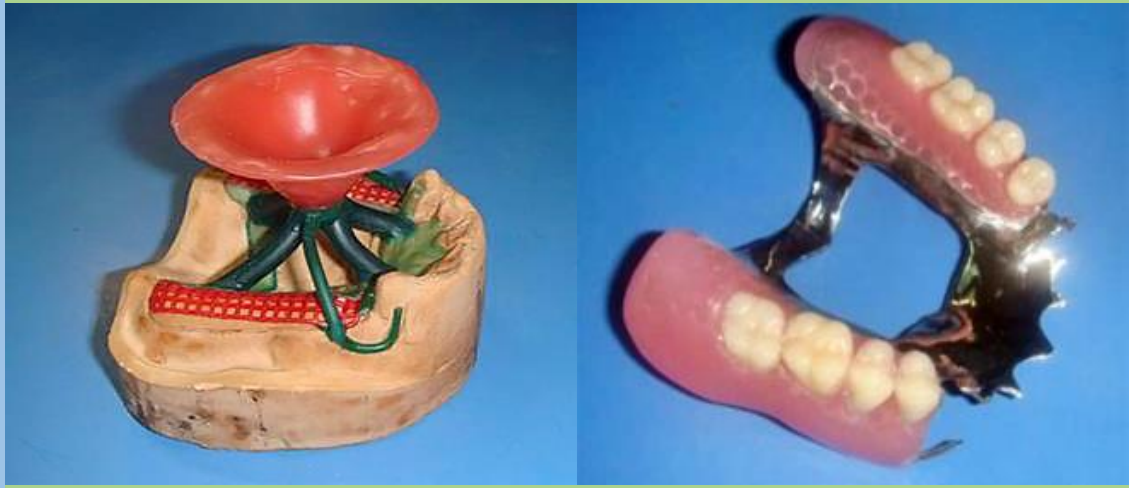


B-Casting Wax:

- These waxes are used to form the wax pattern of the metallic framework of removable partial dentures.
- They are available in sheets and in ready-made shapes

-Many casting waxes possess a slight tackiness to help hold them in position on the gypsum cast before investing and casting.



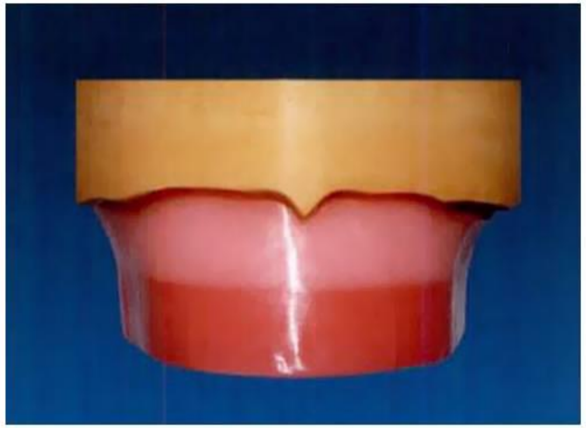


https://mydentaltechnologynotes.wordpress.com/2018/07/09/removable-partial-denture-laboratory-procedures/

- **<u>C-Baseplate Wax</u>**
- Baseplate wax derives its name from its use on the **baseplate of a denture**.
- Baseplates (or record bases) are used to build the contours of denture and hold positions of the denture teeth before processing the denture in acrylic.
- Baseplate wax **is pink or red**, which provides some esthetic quality during making of the denture
- It is available in **sheets 7.5 cm wide, 15 cm long, and 0.13 cm thick.**

- Baseplate wax typically contains **ceresin, beeswax, carnauba wax, and various synthetic waxes.**

https://pocketdentistry.com/dental-waxes-pattern-and-processing-waxes/



https://mydentaltechnologynotes.wordpress.com/2018/06/22/complete-denture-temporary-denture-base-and-occlusal-rims/

- As with all waxes, residual stress can easily be generated and, if released, will move teeth and change the occlusion of the denture

- Steps must be taken to limit these stresses

Important Properties of Pattern Waxes

- 1-All pattern waxes must have low flow at their working temperature (e.g. room temperature) to prevent deformation of the wax pattern.
- 2-The melting range of the wax must be higher than the environment in which the pattern is made.

- 3-On the other hand, too high melting range is undesirable as it may cause development of residual stresses.

- 4-Finally, all pattern waxes must burn out with no residue because the residue would interfere with the casting of the pattern.

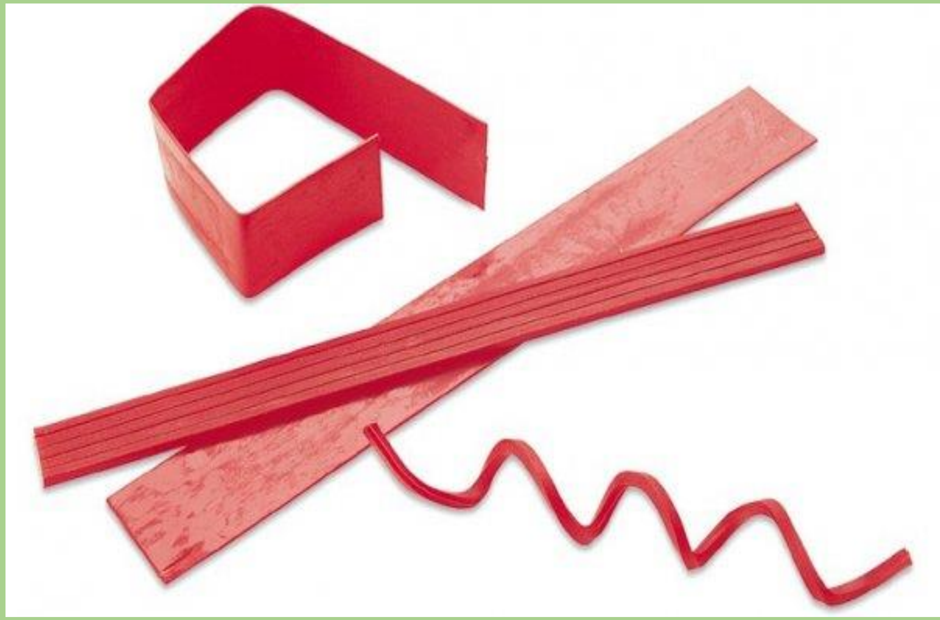
II-PROCESSING WAXES

- Processing waxes have various auxiliary roles in the fabrication of models and impressions and in soldering

1- Boxing and Utility Waxes:

- Boxing and utility waxes are **soft, pliable waxes** used mainly in taking and pouring impressions , have high flow at room temp.

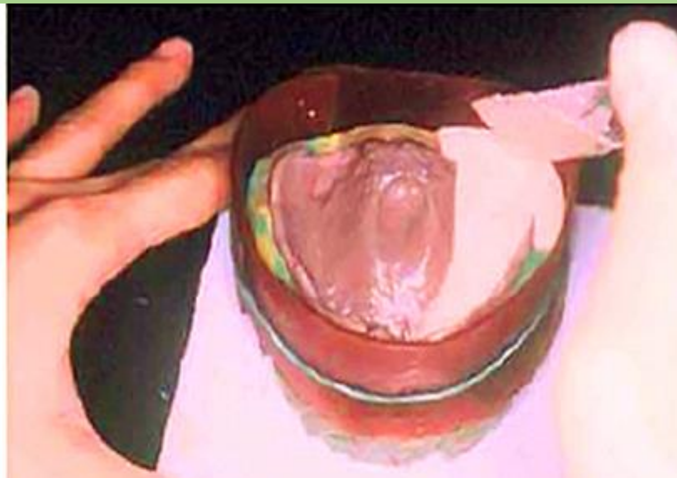
- The waxes are usually **dark in color and have some tackiness, which allows attaching to each other or to gypsum models or to impression trays.**



- **Components:** primarily from **beeswax, paraffin, and other soft waxes**

- **Boxing wax** :supplied as **long (40 cm) strips that are 4 to 5 cm wide and 0.1 cm thick** , molded around an impression to help contain the stone during gypsum pouring when it is vibrated into the impression surface.

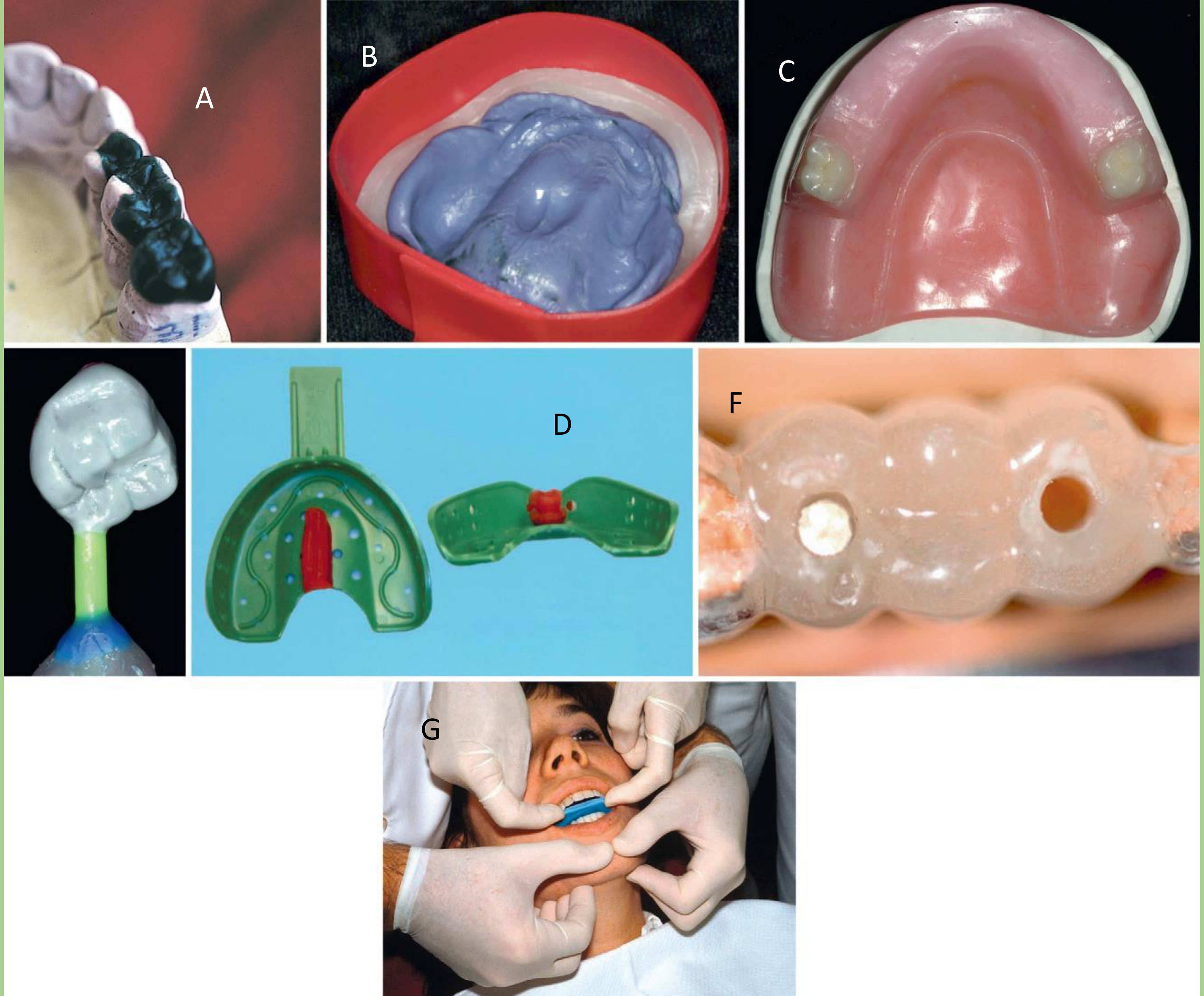
- **Utility and periphery wax are long beads (40 cm or more) about 0.5 cm in diameter**, used with boxing wax during the pouring of impressions. Also, they are used around the periphery of an impression tray to reduce irritation to the soft tissues and/or to extend the tray **before the impression is taken to cover all required teeth and tissues.**



https://mydentaltechnologynotes.wordpress.com/2018/06/29/den
l-waxes/

https://sites.google.com/a/atsu.edu/complete-
dentures/preliminary-impressions

- **<u>2- sticky wax:</u>** usually an orange-colored stick wax, which is <u>*tacky when melted but firm and brittle on cooling*</u> **why? try to guess**
- Sticky waxes are used to temporarily fasten gypsum model components, join and temporarily splint the components of a bridge before process of soldering, or attach pieces of a broken denture before repair.



Applications of waxes in dentistry. A, Inlay pattern wax. B, Boxing wax. C, Baseplate. D, Utility wax. F, Sticky wax. G, Bite wax.

(A, From Hatrick CD, Eakle WS, Bird WF: Dental materials: clinical applications for dental assistants and dental hygienists, ed 2, St Louis, 2011, Saunders; B–D, courtesy Y-W Chen, University of Washington Department of Restorative Dentistry, Seattle, WA; E, From Tsu Y-T: A technique of making impressions on patients with mandibular bony exostoses, J Prosthet Dent 93[4]:400, 2005; F, from Arfai NK, Kiat-Amnuay S: Radiographic and surgical guide for placement of multiple implants, J Prosthet Dent 97[5]:310, 2007; G, from Dawson PE: Functional occlusion: from TMJ to smile design, St Louis, 2007, Mosby.)

REFERENCSES and Photos:

1- Dental Materials applications and foundations 11^{th} ed.

2- Cairo university textbook of Dental Biomaterials


Thank you